

Advait Nair

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EDUCATION

UT Austin

Expected May 2030

Bachelor of Science in Civil Engineering

Austin, TX

- **Coursework:** Civil Engineering Systems; Sequences, Series, and Multivariable Calculus; Principles of Chemistry I

EXPERIENCE

Longhorn Racing Electric

Sep. 2026 – Present

Body Team Member

Austin, TX

- Selected for UT Austin's Formula SAE Electric team, which designs, builds, and races an electric open-wheel vehicle in collegiate competition
- Contributing to the design and build of a torsional rigidity test jig to measure chassis torsional stiffness
- Developed a Python sizing calculator for the jig that derives fixture and stand heights from target floor clearance using rotation geometry and beam deflection, with margin-of-safety checks on roll angle and clearance; verified with 18 unit tests and hand calculations

ShorelineAI

May 2024 – Aug. 2024

Engineering Intern

Remote

- Shoreline AI is a Silicon Valley startup that uses sensors and cloud software to monitor industrial equipment and detect methane leaks at oil and gas sites
- Reverse engineered a wind vane sensor body in CAD from photos and measurements, since no model existed
- Designed a 3D printed mounting bracket that clamps the sensor to a steel crossarm, sizing the clamp and bolts and checking the fit in CAD
- Analyzed environmental conditions in the Permian Basin to inform material and enclosure selection, and delivered the CAD models and drawings to the hardware team

Rahimi Lab, University of Houston

Jun. 2026 – Present

Research Assistant

Houston, TX (Remote)

- Designed and tested evaluation prompts using an "LLM-as-a-judge" methodology to grade the quality of AI-generated questions for a custom LLM
- Audited samples from benchmark datasets containing more than 100,000 question-answer pairs, identifying duplicate-content and data-quality issues that could affect model training.

PROJECTS

NitrogenMAPS Precision Fertilization Concept | *Engineering Design Project - TSA*

- Developed an automated soil-nitrogen monitoring and fertilizer delivery system, pairing an above-ground reservoir with a subsurface distribution network actuated below a target nitrogen threshold
- Benchmarked the concept against nitrification inhibitors and regenerative farming on cost, and proposed a three-level validation plan spanning component, controlled-field, and multi-season trials
- Placed 4th of 45+ teams in Engineering Design at the TSA Texas State Conference (3-person team)

SKILLS

Design & CAD : Solidworks, Onshape, AutoCAD

Programming : Python, Excel, R, HTML/CSS